

CAPSTONE PROJECT REPORT

SDET TRAINING PROGRAM

PARA BANK AUTOMATION TESTING FRAMEWORK

Test Case Analysis Report

ParaBank Banking Application – Automation Testing Project

Project Information

Attribute	Details
Project Name	ParaBank Banking Application Automation Framework
Application	ParaBank
Testing Type	Functional & Negative Automation Testing
Automation Tool	Selenium WebDriver
Framework	TestNG
Language	Java
Design Pattern	Page Object Model (POM)
Reporting Tool	Extent Reports and Allure reports
Logging Framework	Log4j2
CI/CD	Jenkins
Containerization	Docker
Browser Support	Chrome, Safari



Test Execution Summary

Metric	Result
Total Functional Test Cases	10
Total Negative Test Cases	4
Total Automated Scenarios	14
Passed	14
Failed	0
Skipped	0
Pass Percentage	100%

Execution Status

 **PASS**

All planned test cases were executed successfully without any failures.



Functional Test Case Analysis

Test Case ID	Test Scenario	Expected Result	Actual Result	Status
TC_REG_001	Register New Customer	Customer should be registered successfully	Registration successful	PASS
TC_LOGIN_001	Login with Valid Credentials	User should login successfully	Login successful	PASS
TC_LOGIN_002	Invalid Username & Password	Error message should be displayed	Error displayed	PASS
TC_LOGIN_003	Invalid Username & Empty Password	Error message should be displayed	Error displayed	PASS
TC_LOGIN_004	Empty Username &	Error message should be	Error displayed	PASS

Test Case ID	Test Scenario	Expected Result	Actual Result	Status
	Invalid Password	displayed		
TC_ACC_001	Open New Savings Account	New account should be created	Account created successfully	PASS
TC_TRF_001	Transfer Funds	Funds should be transferred successfully	Transfer completed	PASS
TC_BILL_001	Bill Payment	Bill payment should be processed	Payment completed	PASS
TC_FIND_001	Find Transactions	Transaction details should be displayed	Results displayed	PASS
TC_UPDATE_001	Update Contact Information	Profile should be updated	Profile updated	PASS
TC_LOAN_001	Request Loan	Loan request should be processed	Loan processed	PASS
TC_LOGOUT_001	Logout	User should logout successfully	Logout successful	PASS



Negative Test Case Analysis

Defect ID	Scenario	Expected Result	Actual Result	Status
PB-001	Transfer Funds Allows Negative Amount	System should reject negative amount	Transaction accepted	BUG REPRODUCED
PB-002	Bill Payment Allows Negative Amount	System should reject negative amount	Payment accepted	BUG REPRODUCED
PB-003	Same Account Transfer Allowed	System should block transfer	Transfer completed	BUG REPRODUCED
PB-004	Mismatched Account Numbers Accepted	Validation error should be displayed	Payment processed	BUG REPRODUCED



Defect Summary

Defect ID	Severity	Priority	Status
PB-001	Medium	High	Open
PB-002	Medium	High	Open
PB-003	Medium	High	Open
PB-004	Medium	High	Open



Test Coverage Analysis

Coverage Area	Coverage
Functional Testing	100%
UI Validation	95%
Regression Testing	100%
Negative Testing	100%
Cross Browser Testing	50%
Data Driven Testing	100%
Defect Validation Coverage	100%
Reporting & Logging Validation	100%
Overall Automation Coverage	96%



Cross Browser Testing Analysis

Browser Execution Status

Chrome Executed Successfully

Safari Executed Successfully

Firefox Supported (Not Executed)

Observation

The framework executed successfully on Chrome and Safari browsers without any functional differences.



Reporting and Logging Validation

The framework successfully generated:

Reports

- Extent Report
- Allure Report
- TestNG Report
- Execution Summary

Logs

- Log4j2 Execution Logs
- Browser Launch Logs
- Test Start and End Logs

Screenshots

- Automatic Screenshot Capture on Failure



Key Observations

Functional Testing

- All banking workflows executed successfully.
- Registration, Login, Account Creation, Fund Transfer, Bill Payment, Find Transactions, Update Contact Information, Loan Processing, and Logout functionality behaved as expected.

Negative Testing

- Four critical business defects were successfully reproduced through automation.

Framework Validation

- Data Driven Testing worked successfully using CSV files.
 - Extent Reports generated successfully.
 - Jenkins integration executed successfully.
 - Docker execution completed successfully.
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Conclusion

The ParaBank Banking Application Automation Framework was successfully executed and validated.

Final Results

Metric	Result
Total Automated Scenarios	14
Passed	14
Failed	0
Pass Percentage	100%
Defects Reproduced	4
Overall Automation Coverage	96%

Final Status

 **SUCCESSFUL**

The automation framework successfully validated all planned functional workflows and reproduced critical business defects while demonstrating stability, maintainability, and scalability for enterprise-level automation testing.